

So far we have made only the first step in this direction. The next step will be the introduction of the concept of a derivative. This will be the subject of discussion in Dialogues Eight and Nine.

Reader: Nevertheless, I would like to note that in the above examples a function was found to be either continuous everywhere over any interval of finite length or discontinuous at a finite number of points. In this sense the local nature of the concept of a discontinuity point is evident. The continuity of the function, however, is always observed over a certain interval.

Author: First, the continuity of a function within an interval does not interfere with the local nature of continuity. A function is continuous over an interval if it is continuous at all points of this interval.

Second, it is not difficult to construct an example in which the number of discontinuity points over an interval of finite length is infinitely large. Let us look, for example, at the following function:

$$y = \begin{cases} 2 & \text{for } x = \pm 1, \pm \frac{1}{2}, \pm \frac{1}{4}, \pm \frac{1}{8}, \pm \frac{1}{16}, \dots \\ x^2 & \text{for all the remaining points of the real line, including } x=0 \end{cases}$$

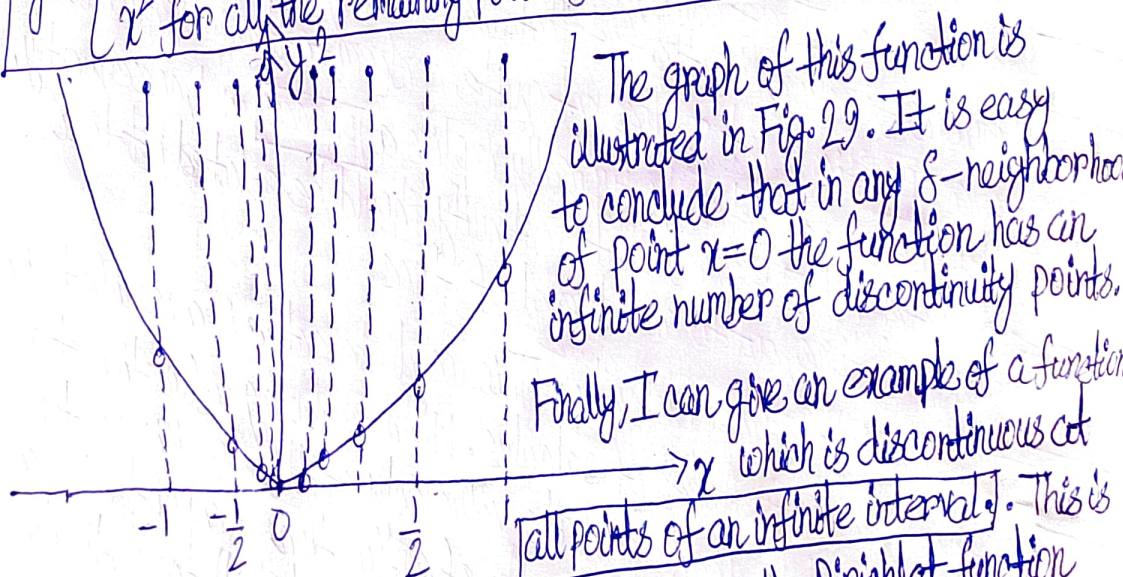


Fig. 29 a function you already know, the Dirichlet function (see the previous dialogue). Being defined on the whole real line, the function has no limit at any point of the real line; consequently, it is discontinuous at each point.

Reader: This is the reason why we in principle cannot plot the Dirichlet function by a graph.